

Estimability, Identifiability and Indicator Variables

When the columns of the model matrix are linearly dependent, the data cannot tell some coefficient vectors apart, because different vectors produce the same mean. This chapter asks which questions about β still have answers. The answer is the same from every direction. A linear function $\lambda^\top \beta$ can be estimated without bias exactly when it is determined by the mean vector $\mathbf{X}\beta$, exactly when λ lies in the row space of $\mathbf{X}$, and exactly when every least squares solution gives it the same value. The rest of the chapter uses this criterion. Reparameterizations and side conditions turn out to be two ways of choosing coordinates, and neither can change what is estimable. Factors enter a regression through indicator variables, and each coding of a factor amounts to a choice of which linear functions of the cell means the coefficients report. Contrasts are the estimable functions of a factor, and orthogonal sets of contrasts split the factor's sum of squares into separate pieces.

What you need. [Chapter 1](#) (rank, generalized inverses and the consistency of linear systems, especially Theorem 1.4.1 and Corollary 1.4.2), [Chapter 2](#) (means and covariances of linear functions, Theorem 2.2.1), [Chapter 4](#) (for the distribution of contrast sums of squares under normality, Theorem 4.4.4), [Chapter 5](#) (the linear model, Definition 5.3.1), and [Chapter 6](#), above all [Section 6.2](#) and [Section 6.4](#), where rank-deficient least squares and the invariance of $\lambda^\top \hat{\beta}$ for $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ (Theorem 6.4.4) first appeared. A few remarks use the Gauss–Markov theorem of [Chapter 7](#).

8.-1.1 Roadmap

- [Identifiability](#)
- [Estimable functions](#)
- [Characterizations of estimability](#)
- [Reparameterization and side conditions](#)
- [Indicator variables and the coding of a factor](#)
- [Several factors, interactions and covariates](#)
- [Contrasts](#)
- [Summary and notes](#)